

Project Proposal

VigilantEye

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Supervisor Meeting Log

Discussion Topic	Task Assignment	Supervisor Signature
Meeting 01: Initial Idea Pitch and Feasibility	Finalize project framework, identify stakeholders (Students, Invigilators, Examination Department, UFM Committee), and define system objectives for AI-based UFM detection and Portal management.	
Meeting 02: Scope Document Finalization	Map out module work division, finalize the technical stack, and draft timeline.	
Meeting 03: UI/UX and Database Architecture Review	Create frontend wireframes and initialize database schema.	
Meeting 04: Core Vision Module Progress Update	Begin DataCollection for UFM detection.	

1. Introduction

Vigilant-Eye: AI Driven UFM Detection and Automated UFM Portal is a smart examination monitoring and case management system designed to improve academic integrity in universities. The system uses AI-enabled CCTV cameras and invigilator reporting to detect unfair means (UFM) activities such as the use of mobile phones, smart watches, electronic devices, or other cheating materials during examinations. When suspicious activity is detected by the AI system, an automatic alert is sent to the monitoring screen in the HOD office/ IT office, and the related evidence video clip is securely stored. If an invigilator identifies a violation, evidence can be uploaded manually through the portal. The system provides dedicated dashboards for Invigilators, Examination Departments, UFM Committees, and Students. It supports complete UFM case management Portal, including case creation, evidence submission, automated notifications, result holding, transcript blocking, case review, decision making, and progress tracking. By automating the entire UFM process, Vigilant-Eye helps universities ensure transparency, efficiency, accountability, and fair examination practices while reducing manual paperwork and delays.

2. Related System Analysis / Literature Review

Application Type / Name	Weakness	Proposed Project Solution (VigilantEye)
Online Proctoring Systems (ProctorU, Honorlock, Examsoft)	They don't provide physical object detection (phones/notes), and have high cloud dependency and privacy concerns, no institutional UFM workflow.	Utilizes deep learning and MediaPipe to capture real-time object + behavior detection, and a complete UFM case management portal for institutional escalation.
Paper-Based UFM Reporting Process	UFM cases are documented manually, leading to paperwork delays, incomplete records, and difficulty in tracking case progress.	Provides a digital UFM portal where invigilators can create cases, upload evidence, apply digital signatures, and track case status through their portal.
AI-Based Cheating Detection Systems	Limited to detecting isolated objects or gestures, no real time alerts, and lack institutional workflow integration.	Develops a fully deployable system combining YOLOv8 object detection, MediaPipe pose/gaze tracking, spatial seat mapping, and a structured UFM case management portal for real institutional use.
Existing Academic Integrity Monitoring Approaches	Most systems detect violations but do not automate the entire workflow from detection to final decision and result management.	Delivers an end-to-end solution covering AI-based detection, evidence collection, case creation, workflow automation, decision support, audit trails, and final resolution management.

3. Problem Statement

In the traditional examination systems, maintaining fairness and controlling Unfair Means (UFM) has become increasingly difficult due to large class sizes, human limitations of invigilators, and the lack of real-time monitoring tools. Most universities still rely on manual invigilation and basic CCTV cameras that only record footage without intelligent analysis. As a result, cheating activities such as use of mobile phones, smart watches, handwritten notes, and other electronic devices often go unnoticed during the exam or are detected too late during post-exam review. There is also no proper centralized system to instantly document, manage, and track UFM cases across departments, which leads to delays, missing evidence, and inconsistent decision-making. Therefore, there is a strong need for an intelligent and automated solution that can detect suspicious behavior in real time and immediately convert it into a structured institutional case. The proposed system, VigilantEye, addresses this gap by using AI-based CCTV and invigilator input to detect UFM incidents, automatically capture and store evidence, and generate alerts in real time. It also provides a complete UFM Portal with role-based dashboards for invigilators, examination departments, committees, and students to ensure transparent reporting, fast decision-making, and fully automated case tracking and result handling.

4. Problem Solution / Objectives of the Proposed System

To address the growing challenge of maintaining academic integrity in physical examination halls, this project proposes VigilantEye: AI-Driven UFM Detection and Automated UFM Portal, an intelligent system designed to detect, document, and manage Unfair Means (UFM) cases in real time. Traditional examination monitoring methods rely heavily on manual invigilation and passive CCTV recording, which are unable to provide instant detection, structured evidence handling, or fast institutional response. As a result, many cheating incidents go unnoticed during exams or are reported late without proper evidence and coordination between departments.

The proposed solution introduces an AI-powered monitoring layer that continuously analyzes live CCTV feeds and invigilator inputs to detect suspicious activities such as the use of mobile phones, smart watches, electronic gadgets, hidden notes, and peer interaction. When a potential UFM event is detected, the system automatically generates an alert on the invigilator and HOD monitoring dashboard and simultaneously captures and stores a short evidence video clip with timestamped metadata. If an invigilator manually identifies an incident, they can also instantly record evidence and trigger the same automated workflow.

Alongside detection, the system includes a fully integrated UFM Portal that transforms incidents into structured institutional cases. The portal ensures seamless communication between invigilators, the examination department, and the UFM committee through role-based dashboards, automated notifications, and centralized case tracking. This eliminates delays, reduces paperwork, and ensures transparency in decision-making.

4.1 Objectives

The primary objective of VigilantEye is to strengthen academic integrity by combining AI-based examination monitoring with a centralized digital UFM management platform. The system aims to reduce manual effort, improve detection accuracy, and streamline the entire UFM case handling process.

1. Ai-based Exam Rooms Surveillance

Traditional examination monitoring depends on invigilators manually observing multiple students and CCTV screens, making it difficult to continuously detect all suspicious activities. This project aims to enhance

examination integrity by implementing an AI-based CCTV surveillance system that continuously monitors examination halls, detects suspicious activities and prohibited objects, records evidence clips, generates real-time alerts, and assists invigilators in identifying potential UFM incidents efficiently.

2. Automated UFM Case Management Portal

Most institutions still use manual reporting procedures, resulting in delayed processing, incomplete records, and limited case visibility. This objective aims to digitize and automate the complete UFM reporting and investigation process through a centralized portal that enables evidence management, workflow automation, case tracking, and seamless communication among all stakeholders, including Invigilators, HODs, DEC's, Examination Departments, UFM Committees, and Students.

3. Institutional Transparency and Administrative Efficiency

To create a centralized institutional platform that improves transparency, accountability, and administrative efficiency in UFM case handling through automated workflows, structured case management, and seamless stakeholder coordination..response.

5. Scope

The scope of this project includes the design, development, and deployment of a real-time intelligent surveillance and institutional UFM management Portal named VigilantEye: AI-Driven UFM Detection and Automated UFM Portal. The system is intended for use in physical examination halls across universities and educational institutions, where maintaining academic integrity and controlling Unfair Means (UFM) is critical. The solution integrates AI-based video analytics with a centralized web-based portal to ensure continuous monitoring, instant detection, and structured handling of examination violations. The system will be deployed in controlled institutional environments such as examination control rooms, HOD offices, and examination departments, where live CCTV feeds and invigilator inputs are monitored in real time. It is designed specifically for multi-candidate physical examination settings and is not intended for remote or online examination use cases. The system operates within academic examination workflows and supports complete UFM lifecycle management from detection to final committee decision.

VigilantEye requires existing CCTV infrastructure or IP cameras and standard computing hardware for running AI inference pipelines. It is designed to be scalable across single or multiple examination halls, ensuring consistent monitoring regardless of hall size or candidate density. The system emphasizes accessibility, institutional integration, and real-time response without relying on external cloud-based surveillance services.

5.1 In-Scope Functionalities:

The core technical contribution of this project is a real-time AI-driven UFM detection and institutional case management system powered by computer vision models (YOLO-based object detection) and structured workflow automation through a centralized UFM portal. The system supports continuous monitoring of physical examination halls through live CCTV/IP camera feeds installed in control rooms or HOD monitoring offices, where suspicious activities are detected and processed in real time.

A repeated error validation mechanism is included to reduce false positives by confirming suspicious activity across multiple frames before triggering a final alert and evidence capture.

Beyond the core AI engine, the scope includes a complete user-facing ecosystem comprising:

- **Role-based user authentication** and dashboard management for Invigilators, Examination Department, UFM Committee, and Students.
- **UFM case creation** module allowing invigilators to generate structured digital cases with student information and exam details (ID, name, department, date, time, room, and camera ID).
- **Automated evidence capture system** that stores short video clips, snapshots, timestamps, and camera metadata during detected incidents.
- **Digital UFM form submission** with fields for violation type, incident description, remarks, and evidence attachment.
- **Instant notification system (portal + email)** to alert Examination Department and UFM Committee upon case submission.
- **Examination Department dashboard** for monitoring total cases, pending investigations, department-wise statistics, and semester-wise UFM trends.
- **Automated result control system** including result hold, transcript blocking, audit trail generation, and controlled result release after final decision.
- **UFM Committee review system** for evidence analysis, student history evaluation, case assignment, and final decision approval.

6. Modules

The VigilantEye system is organized into six interconnected modules that collectively support the detection, reporting, management, and resolution of Unfair Means (UFM) incidents. These modules work together to provide real-time examination monitoring, automated evidence collection, structured case management, stakeholder coordination, and transparent decision-making. The modular architecture ensures scalability, maintainability, and efficient operation across all stages of the UFM lifecycle, from incident detection to final case resolution.

6.1 Real-Time UFM Detection & Surveillance Engine (Core Module)

This is the foundational module of VigilantEye and represents the core intelligence layer of the system. The module processes live CCTV/IP camera streams installed in examination halls and continuously analyzes video frames in real time using AI-based computer vision models (e.g., YOLOv8 object detection and behavior analysis algorithms). The system is designed specifically for physical examination environments where multiple candidates are monitored simultaneously across a hall layout. It identifies potential Unfair Means (UFM) activities such as mobile phone usage, smart watches, electronic gadgets, hidden notes, paper exchange, and suspicious hand or head movements that indicate cheating behavior.

1. When a suspicious object or behavior is detected, the system immediately triggers a real-time alert on the Invigilator and HOD monitoring dashboard.
2. The module continuously tracks candidate activity across frames to reduce false positives by validating detection consistency before confirming a UFM event.
3. Each detection event is assigned a confidence score and timestamp for traceability and audit purposes.

press.

6.2 Automated Evidence Capture & Incident Logging

This module ensures that every detected UFM event is automatically converted into verifiable digital evidence. When a violation is confirmed, the system automatically extracts a short video clip and key snapshot frames from the live CCTV feed. Each evidence package includes metadata such as camera ID, room number, seat location (if mapped), timestamp, and detection confidence level. Evidence is securely stored in the institutional database and linked directly to the corresponding UFM case. The system also supports manual evidence upload by invigilators in case of human-detected incidents.

6.3 UFM Case Creation & Management Module

This module provides a structured workflow for creating, submitting, and managing UFM cases within the institution. It standardizes incident reporting through digital forms and ensures all required information is captured consistently.

- Auto-fills student details (ID, name, department, exam info).
- Allows invigilator to select violation type and add remarks.
- Attaches AI-generated or manually uploaded evidence.
- Requires digital signature from Invigilator and HOD.
- Submits structured UFM case to Examination Department automatically.

6.4 Role-Based Dashboard System

This module provides separate dashboards for all stakeholders involved in the UFM process, ensuring controlled access to relevant information and functionalities based on user roles. The system enhances transparency, accountability, and coordination throughout the UFM case lifecycle by enabling structured communication and workflow management among Invigilators, Heads of Departments (HODs), Departmental Examination Committees (DECs), Examination Departments, UFM Committees, and Students.

- **Invigilator Dashboard:** Create UFM cases, upload evidence, add incident remarks, submit cases for verification, and track case status throughout the workflow.
- **HOD (Head of Department) Dashboard:** Review cases submitted by invigilators, verify evidence, add departmental remarks, approve or return cases, and forward verified cases to the Departmental Examination Committee (DEC).
- **Departmental Examination Committee (DEC) Dashboard:** Review forwarded cases, analyze evidence and reports, record committee recommendations, conduct preliminary investigations, and forward cases to the Examination Department.
- **Examination Department Dashboard:** View all registered cases, monitor institution-wide statistics, track pending investigations, manage result hold and transcript restrictions, maintain records, and forward cases to the UFM Committee.
- **UFM Committee Dashboard:** Review evidence, analyze student history, evaluate cases, conduct final reviews, record decisions, issue disciplinary actions, and generate official decision notifications.
- **Student Dashboard:** View UFM case details, track case progress, receive notifications, view case outcomes, and submit remarks, explanations, or clarifications.

6.5 Automated Notification & Alerting System

This module ensures real-time communication and coordination between all stakeholders involved in the UFM lifecycle. It plays a critical role in maintaining fast response and transparency during examination monitoring. Whenever a suspicious activity or UFM event is detected, instant alerts are generated and displayed on the Invigilator and HOD dashboards to ensure immediate attention. In addition, once a UFM case is created and submitted, the system automatically sends email and portal notifications to the Examination Department and UFM Committee for prompt action. The module also provides continuous status updates throughout the case workflow, keeping all relevant users informed at every stage such as Pending, Under Review, Approved, or Rejected. This ensures that no case progresses without proper visibility, and all stakeholders remain synchronized in real time.

6.6 Audit Trail & Compliance Logging Module

This module is a core governance and accountability layer of the VigilantEye system and is responsible for ensuring complete transparency, traceability, and integrity across all Unfair Means (UFM) operations. It continuously records and maintains a secure log of every action performed within the system, ensuring that all activities related to case handling, evidence processing, and decision-making are permanently stored for audit and compliance purposes. The module is designed to support institutional accountability by providing a tamper-proof activity history that can be reviewed at any time by authorized authorities.

1. **Activity Logging Structure:** The audit system captures and logs every system-level and user-level action in real time. This includes UFM case creation, modifications to case details, evidence uploads, case assignments, committee reviews, and final decision entries. Each log entry is time-stamped and associated with a unique user ID and role (Invigilator, Examination Department, Committee Member, or Student).
2. **Secure & Tamper-Proof Storage:** All logs are stored in a secure database with restricted access controls to ensure that records cannot be altered or deleted without proper authorization, maintaining data integrity and legal reliability.
3. **Cross-Dashboard Tracking:** The module tracks user activity across all dashboards, ensuring end-to-end visibility of the entire UFM lifecycle from detection to final resolution.
4. **Audit Reporting Support:** The system can generate structured audit reports summarizing case histories, user actions, and institutional responses for internal review and regulatory compliance.
5. **Transparency & Accountability Enforcement:** By maintaining a complete digital footprint of all actions, the module ensures fairness, prevents manipulation, and strengthens trust in the decision-making process of the institution.

7. Data Gathering Approach

To ensure accurate and reliable detection of Unfair Means (UFM) in physical examination halls, the system will utilize publicly available computer vision datasets along with custom-collected CCTV footage from controlled laboratory environments. Pre-trained object detection models such as YOLO will be fine-tuned on datasets containing relevant objects like mobile phones, smart watches, electronic devices, and paper-based cheating materials. Additionally, synthetic and real-world annotated video samples of examination hall scenarios will be used to train and validate behavior detection and alert generation modules.

For system enhancement and operational modules, examination-related metadata such as seating plans, student enrollment data, and hall layouts will be integrated through institutional databases. The UFM Portal will rely on structured form-based data entry from invigilators and examination departments to continuously enrich the system’s case history database. Continuous testing in simulated exam hall environments and feedback from invigilators will be used to refine detection accuracy, reduce false positives, and improve the efficiency of the alerting and case management workflow.

8. Tools and Technologies

Tool / Technology	Purpose / Rationale
React.js + Tailwind CSS	Web-based monitoring and case management dashboard
FastAPI	High-performance backend and real-time communication
PostgreSQL	Storage of users, UFM cases, evidence, and audit logs
OpenCV	Video processing and image enhancement
YOLOv8	AI-based UFM object detection
MediaPipe	Candidate behavior and pose analysis
WebSocket	Instant alert and notification system
RTSP	Live CCTV/IP camera connectivity
Python / Deep Learning Models	Used to train custom classification/regression models that analyze the angles derived from MediaPipe to determine if a student posture is "Correct" or "Incorrect".
Docker, Git, Postman, VS Code	Development, testing, deployment, and version control tools

9. Project Stakeholders and Roles

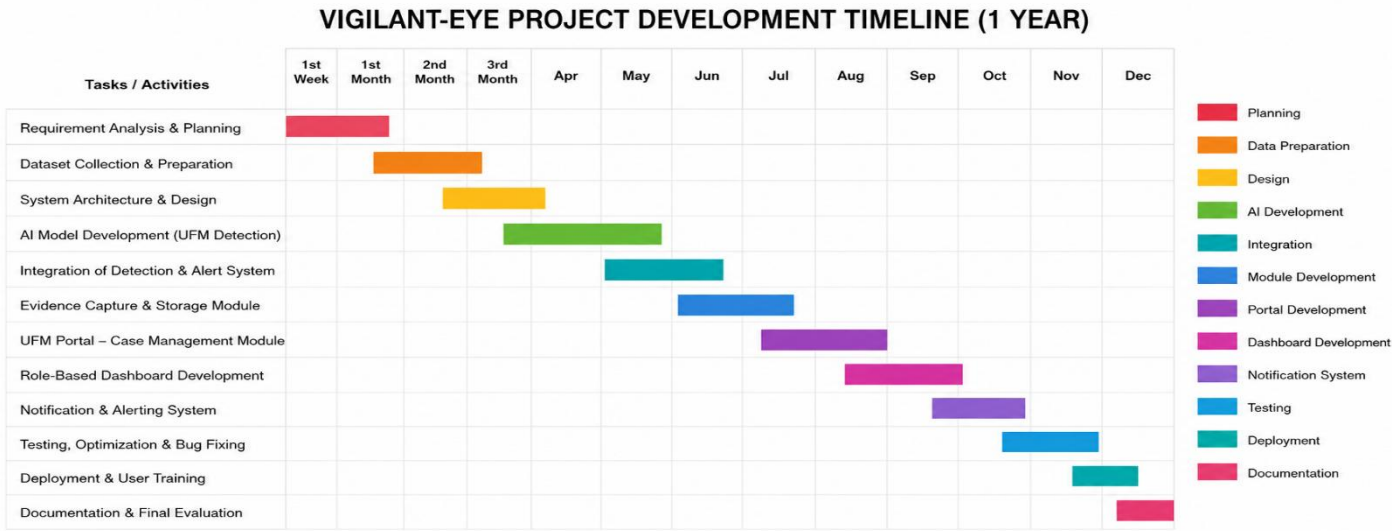
Stakeholder	Role in Project
Air University Islamabad	Project Sponsor / Institution
Dr. Muhammad Bilal Khan	Project Supervisor (Guiding technical and academic progress)
FYP Evaluation Committee	Final Evaluation and Grading

Muhammad Abdullah	AI & Computer Vision Developer
Muhammad Talha	Backend Developer
Muzamil Wasif	Frontend & UI/UX Developer

10. Module-based Work Division

Team Member	Domain	Assigned Responsibilities
Muhammad Abdullah 232514	AI/ML Core	- YOLOv8 model fine-tuning - Suspicion score matrix - FastAPI backend, alert routing engine - Database schema design
Muhammad Talha 232950	Frontend/UX	- React + Tailwind dashboard - live camera feed rendering - WebSocket client integration - Alert notification panel - Seat map visualization
Muzamil Wasif 232430	CV & System Integration	- OpenCV frame pre-processing pipeline - MediaPipe head pose & gaze estimation - RTSP camera feed management - System integration testing - Evidence capture module

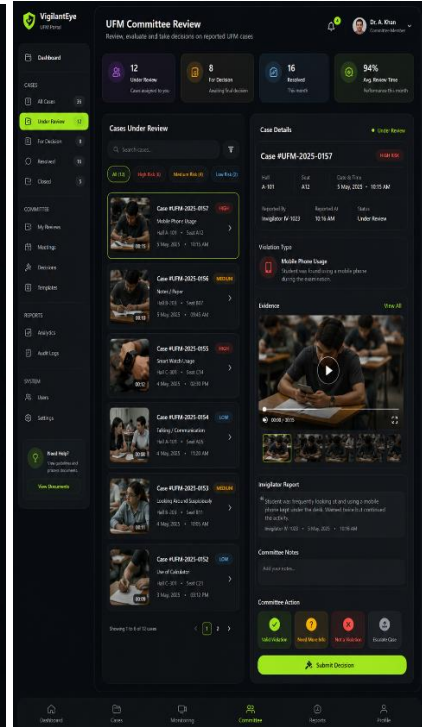
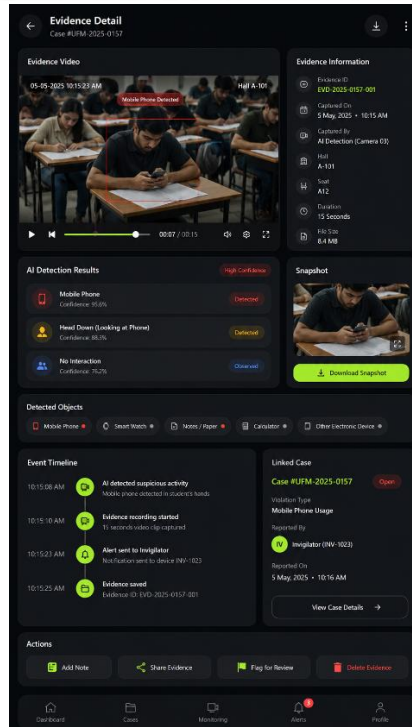
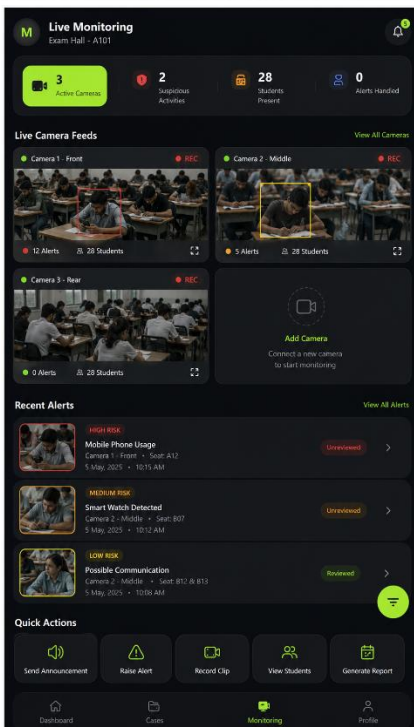
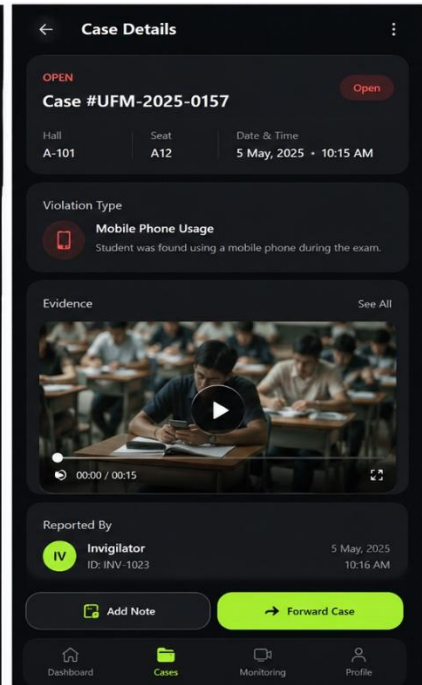
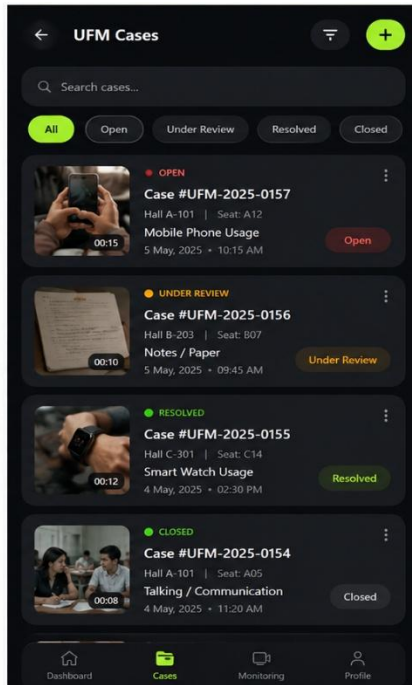
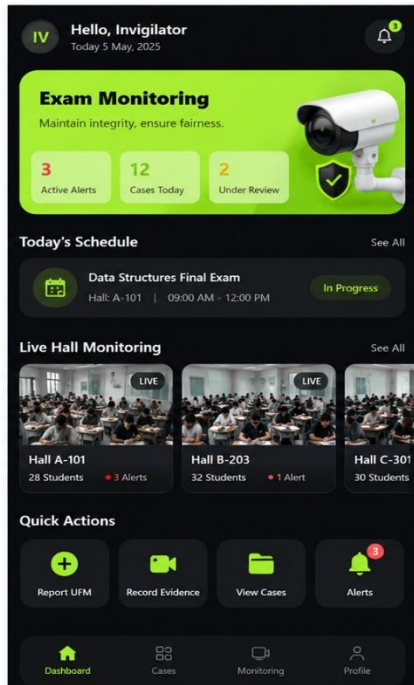
11. Gantt Chart (Project Timeline)



12. Mockups

Invigilator Dashboard

UFM Case Page



13. References

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